

RA-04: Clustered-gap bounds for randomized block Krylov approximation

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Abstract

A complete proof of the displayed RA-04 assertion is claimed. For a positive head spectrum of size $m = bt$ whose relative b -step gap is at least Δ , the tangential interpolation evaluation matrix satisfies

$$\mathbb{P}\left\{\sup_{0 \leq x \leq \lambda_m} \|E_{\lambda, H}(x)\|_2 \leq 2^{24} m^8 \eta^{-4} \left(\frac{128e}{\Delta}\right)^t\right\} \geq 1 - \eta.$$

The proof combines a logarithmic root covering, an annular resolvent identity, and a variational descent estimate. A published real-algebraic tube theorem controls proximity to the singular head-Krylov locus. Its defining polynomial has degree m , independently of the spectral scale. Translation preserves that locus, so one probability event controls every tail evaluation. The standard good-start convergence theorem then gives the exact iteration order requested in RA-04, without an additional $t \log m$ term.

Review status. The full exact RA-04 target passed a separate Codex AI-agent informal mathematical audit on 12 September 2026 (America/New_York). The repository submission proposes *Solved* under its independent-audit policy. This is not external human peer review or formal verification. No Lean verification was performed. The independent report is included in `verification/independent-review.md`. Finite tests are diagnostics, not proof certificates.

1 The assertion and the new estimate

Let $A \in \mathbb{R}^{n \times d}$ and $M = AA^T$. Order the eigenvalues of M as $\lambda_1 \geq \dots \geq \lambda_n \geq 0$, extending this sequence by zeros for indices greater than n . Fix

$$1 \leq b \leq k, \quad t = \lceil k/b \rceil, \quad m = bt \leq \text{rank}(A).$$

The relative clustered gap is

$$\Delta = \min_{1 \leq i \leq m-b} \frac{\lambda_i - \lambda_{i+b}}{\lambda_i} > 0. \quad (1.1)$$

When $m = b$, use the empty-minimum convention $\Delta = 1$. All logarithms are natural. All assertions concern exact arithmetic and the original, unperturbed matrix A .

For a standard real Gaussian $G \in \mathbb{R}^{n \times b}$, define

$$\mathcal{K}_q(M, G) = \text{range}[G, MG, \dots, M^{q-1}G], \quad \hat{A} = Z[Z^T A]_k, \quad (1.2)$$

where Z is any orthonormal basis of this space and $[\cdot]_k$ is a truncated SVD. These are precisely the algorithmic conventions of RA-04 [1].

Theorem 1.1 (RA-04). *There is an absolute constant $C > 0$ such that, for every input above and every $0 < \varepsilon, \delta < 1/2$, choosing*

$$q = \left\lceil \frac{C}{\sqrt{\varepsilon}} \left(t \log \frac{2}{\Delta} + \log \frac{n}{\delta \varepsilon} \right) \right\rceil \quad (1.3)$$

in (1.2) gives, with probability at least $1 - \delta$,

$$\|A - \hat{A}\|_{\xi} \leq (1 + \varepsilon) \|A - [A]_k\|_{\xi}, \quad \xi \in \{2, F\}, \quad (1.4)$$

$$|\|Av_i\|_2^2 - \lambda_i| \leq \varepsilon \lambda_{k+1}, \quad 1 \leq i \leq k, \quad (1.5)$$

where v_i are the ordered top k right singular vectors of $\hat{A}$. The zero-optimal-error case is included.

The earlier partial reports proved an all-input estimate with $t \log(2m/\Delta)$ in place of $t \log(2/\Delta)$. They left a dimension-polynomial, gap-exponential interpolation bound unproved. The present argument proves that bound directly. It does not rely on the earlier restricted-width cluster theorem or on the earlier fractional-moment recurrence.

The two substantive external inputs are explicitly isolated: the real-algebraic conic tube theorem in Section 6, and the standard deterministic Krylov good-start theorem in Section 7. The root-covering, contour, descent, and interpolation arguments are proved below. None uses an unproved small-ball or interpolation premise.

2 Head interpolation and its singular locus

For the moment, work only with positive nodes $\lambda_1 \geq \dots \geq \lambda_m > 0$, where $m = bt$ and (1.1) holds. Write $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m)$ and, for any $H \in \mathbb{R}^{m \times b}$, put

$$K_{\lambda}(H) = [H, \Lambda H, \dots, \Lambda^{t-1} H] \in \mathbb{R}^{m \times m}. \quad (2.1)$$

Its rows are $h_i^T [I_b, \lambda_i I_b, \dots, \lambda_i^{t-1} I_b]$. Whenever it is invertible, set

$$E_{\lambda, H}(x) = [I_b, x I_b, \dots, x^{t-1} I_b] K_{\lambda}(H)^{-1}. \quad (2.2)$$

For $y \in \mathbb{R}^m$, $P(x) = E_{\lambda, H}(x)y$ is the unique vector polynomial of degree at most $t-1$ satisfying $h_i^T P(\lambda_i) = y_i$.

Lemma 2.1 (Generic invertibility). *The polynomial $D_{\lambda}(H) = \det K_{\lambda}(H)$ is nonzero and homogeneous of degree m in the mb entries of H . Consequently a standard Gaussian H gives an invertible $K_{\lambda}(H)$ almost surely.*

Proof. Every entry of $K_{\lambda}(H)$ is linear in H , so its determinant is homogeneous of degree m unless identically zero. To exclude that possibility, set row i of H equal to $e_{1+((i-1) \bmod b)}^T$. Row and column permutations then give b scalar Vandermonde blocks. The nodes in block r are $\lambda_r, \lambda_{r+b}, \dots, \lambda_{r+(t-1)b}$, which are distinct by (1.1). Thus all blocks are invertible. For $t = 1$, this witness is simply $H = I_b$. A nonzero real polynomial vanishes on a Lebesgue-null set, proving the Gaussian assertion. $\square$

Define the closed algebraic cone

$$\Sigma_{\lambda} = \{H \in \mathbb{R}^{m \times b} : D_{\lambda}(H) = 0\}. \quad (2.3)$$

The distance to this cone is always measured in the Frobenius norm on the *starting matrix* H , not in an unrestricted perturbation norm on the raw matrix $K_{\lambda}(H)$.

Lemma 2.2 (Translation invariance). *For every real x ,*

$$\Sigma_{\lambda-x} = \Sigma_{\lambda}, \quad E_{\lambda-x,H}(0) = E_{\lambda,H}(x) \quad (H \notin \Sigma_{\lambda}). \quad (2.4)$$

In fact, $D_{\lambda-x}(H) = D_{\lambda}(H)$ for every H .

Proof. Expanding $(\Lambda - xI)^j H$ gives $K_{\lambda-x}(H) = K_{\lambda}(H)R_x$, where R_x is block upper triangular with (ℓ, j) block $\binom{j}{\ell}(-x)^{j-\ell}I_b$ for $0 \leq \ell \leq j \leq t-1$. All diagonal blocks are I_b , so $\det R_x = 1$. This proves the determinant and locus assertions. Translating a vector interpolant as $P(y) \mapsto P(y+x)$ proves the evaluation identity by uniqueness. $\square$

This exact invariance will be important: the probability event used below is unchanged by every common shift of the nodes. A union bound over evaluation points is unnecessary.

3 A logarithmic root covering

The next elementary covering has total length proportional to h , not to the degree. That distinction prevents a $t \log t$ loss.

Lemma 3.1 (One-dimensional root covering). *Let $a_1, \dots, a_s \in \mathbb{R}$, with multiplicities allowed, and let $0 < h \leq 1$. There is a finite union V of closed intervals, of total length at most $6h$, such that*

$$\prod_{\ell=1}^s \min\{1, |y - a_{\ell}|\} \geq (h/e)^s \quad (3.1)$$

whenever y is outside the interior of V . For $s = 0$, take $V = \emptyset$ and the empty product equal to one.

Proof. For $s > 0$, let $d_j(y)$ be the j th smallest value of $|y - a_{\ell}|$ and define the open bad set

$$\Omega = \{y : d_j(y) < jh/s \text{ for at least one } j \in \{1, \dots, s\}\}.$$

For each such pair (y, j) , the open interval $I(y, j)$ centered at y with radius jh/s contains at least j of the points, counted with multiplicity.

From this family, select an interval of largest available length, remove all intervals intersecting it, and repeat. There are only s possible lengths. Selected intervals are disjoint, and each contains at least one point in its interior, so at most s selections occur. If j_{ℓ} corresponds to a selected interval, disjointness gives $\sum_{\ell} j_{\ell} \leq s$. Hence the sum of the selected lengths is at most $2h$.

Every discarded interval meets a selected interval at least as long. Its center is therefore at distance less than twice the radius of that selected interval. Consequently Ω lies inside the interior of the union V of the closed triple dilates of the selected intervals. This union has total length at most $6h$.

Outside Ω , $d_j(y) \geq jh/s$. Since $jh/s \leq 1$,

$$\prod_{\ell=1}^s \min\{1, |y - a_{\ell}|\} \geq \prod_{j=1}^s \frac{jh}{s} = h^s \frac{s!}{s^s} \geq (h/e)^s.$$

The last inequality follows from $\sum_{j=1}^s \log j \geq \int_1^s \log u \, du \geq s \log s - s$. Every point outside the interior of V is outside Ω . $\square$

Corollary 3.2 (Radial lower bound). *Let ϕ be a real or complex scalar polynomial of degree s with $\phi(0) = 1$. For $0 < h \leq 1$ there is a union $V \subset \mathbb{R}$ of closed intervals of total length at most $6h$ such that*

$$|\phi(z)| \geq \left(\frac{h}{2e}\right)^s \quad \text{whenever } \log |z| \notin \text{int } V. \quad (3.2)$$

Proof. The assertion is immediate for $s = 0$. Otherwise write $\phi(z) = \prod_{\ell=1}^s (1 - z/\zeta_\ell)$, where every $\zeta_\ell \neq 0$ because $\phi(0) = 1$. Apply Lemma 3.1 to $a_\ell = \log |\zeta_\ell|$. If $y = \log |z|$, the reverse triangle inequality gives

$$|1 - z/\zeta_\ell| \geq |1 - e^{y-a_\ell}| \geq \frac{1}{2} \min\{1, |y - a_\ell|\}.$$

For a nonnegative exponent, use $e^u - 1 \geq u$; for a negative exponent, use $1 - e^{-u} \geq (1 - e^{-1}) \min\{1, u\} \geq \frac{1}{2} \min\{1, u\}$. Multiplication and (3.1) prove the result. $\square$

4 The annular moment estimate

For a row set J , H_J denotes the corresponding submatrix. If $|J| \leq b$, the notation $\sigma_{\min}(H_J)$ means its smallest row singular value, equivalently

$$\|H_J^T z\|_2 \geq \sigma_{\min}(H_J) \|z\|_2 \quad (z \in \mathbb{R}^{|J|}).$$

If every contiguous b -row submatrix has smallest singular value at least μ , then the same holds for every smaller contiguous row set: extend that set to b contiguous rows and pad z with zeros.

Lemma 4.1 (Paired nondegeneracy). *Suppose $H \in \mathbb{R}^{m \times b}$ satisfies $\|H\|_2 \leq R$ and every contiguous b -row submatrix has smallest singular value at least $\mu > 0$. Let $u \in \mathbb{R}^m$ satisfy*

$$\|u\|_2 = 1, \quad H^T \Lambda^j u = 0 \quad (1 \leq j \leq t-1). \quad (4.1)$$

For every scalar polynomial ϕ of degree at most $t-1$ with $\phi(0) = 1$,

$$\|\text{diag}(\phi(\lambda_1), \dots, \phi(\lambda_m))u\|_2 \geq \frac{\mu\gamma\rho}{4\sqrt{m}R}, \quad \gamma = \frac{\Delta}{64m}, \quad \rho = \left(\frac{\Delta}{128e}\right)^{t-1}. \quad (4.2)$$

No invertibility of $K_\lambda(H)$ is assumed in this lemma.

Proof. Use Corollary 3.2 with $h = \Delta/64$. Since $\deg \phi \leq t-1$, the resulting lower bound on its allowed circles is at least ρ . Enlarge its covering set to

$$U = V \cup \bigcup_{i=1}^m [\log \lambda_i - \gamma, \log \lambda_i + \gamma]. \quad (4.3)$$

The total length of U is at most

$$6\frac{\Delta}{64} + 2m\frac{\Delta}{64m} = \frac{\Delta}{8}.$$

Every connected component therefore has length at most $\Delta/8$. It contains at most b nodes $\log \lambda_i$, counting multiplicities. Indeed, $b+1$ such nodes would include some i and $i+b$ and give

$$\log \lambda_i - \log \lambda_{i+b} \leq \Delta/8,$$

whereas (1.1) gives this difference at least $-\log(1 - \Delta) \geq \Delta$. When $t = 1$, there are only b nodes in total, so this conclusion is immediate.

Consider a component $[a, c]$ containing at least one node. Its node indices form a contiguous set J of size at most b . Both endpoints are outside the interior of V , and their distance from every $\log \lambda_i$ is at least γ : an endpoint at a smaller distance would be interior to a node interval in (4.3). Thus, on each of the circles $|z| = e^a$ and $|z| = e^c$,

$$|\phi(z)| \geq \rho, \quad \max_i \frac{\lambda_i}{|z - \lambda_i|} \leq \frac{1}{1 - e^{-\gamma}} \leq \frac{2}{\gamma}. \quad (4.4)$$

The last inequality uses $0 < \gamma \leq 1$.

Put $w = \text{diag}(\phi(\lambda_i))u$. For any z outside the nodes, the function of x

$$r_z(x) = x \frac{\phi(x) - \phi(z)}{x - z}$$

is a polynomial of degree at most $t - 1$ with zero constant coefficient. The moment equations (4.1) imply

$$\phi(z) H^T \Lambda(zI - \Lambda)^{-1} u = H^T \Lambda(zI - \Lambda)^{-1} w. \quad (4.5)$$

Let Γ_J consist of the outer circle $|z| = e^c$ traversed counterclockwise and the inner circle $|z| = e^a$ traversed clockwise. Divide (4.5) by $z\phi(z)$ and integrate:

$$H_J^T u_J = \frac{1}{2\pi i} \int_{\Gamma_J} \frac{H^T \Lambda(zI - \Lambda)^{-1} w}{z\phi(z)} dz. \quad (4.6)$$

To verify the left side, integrate the equivalent left integrand $H^T \Lambda(zI - \Lambda)^{-1} u/z$. For each coordinate,

$$\frac{\lambda_i}{z(z - \lambda_i)} = \frac{1}{z - \lambda_i} - \frac{1}{z}.$$

The two circles enclose exactly the head nodes indexed by J in their annulus; their contributions from zero cancel. Repeated nodes give the sum of their row contributions. Zeros of ϕ inside the annulus cause no difficulty: the identity is used on the boundary, and residues are evaluated only for the left integrand, which contains no $1/\phi$.

On either circle, its length cancels the factor $1/|z|$. Consequently (4.4) and (4.6) give

$$\|H_J^T u_J\|_2 \leq \frac{4R}{\gamma\rho} \|w\|_2, \quad \|u_J\|_2 \leq \frac{4R}{\mu\gamma\rho} \|w\|_2.$$

The components containing nodes partition all m indices, and there are at most m of them. Summing the squared inequalities yields

$$1 = \|u\|_2^2 \leq m \left(\frac{4R}{\mu\gamma\rho} \right)^2 \|w\|_2^2,$$

which proves (4.2). □

The proof uses at most b rows at a time, but does not fix a partition into b -sized spectral clusters. It therefore allows arbitrary nonzero-width clusters, exact repeated values, and unaligned spectra subject only to (1.1).

5 Extrapolation, descent, and distance to singularity

For $H \notin \Sigma_\lambda$, define

$$f_\lambda(H) = \frac{1}{\|E_{\lambda,H}(0)\|_2}. \quad (5.1)$$

This is positive and continuous on the open set $\mathbb{R}^{m \times b} \setminus \Sigma_\lambda$.

Lemma 5.1 (Variational formula and a descent direction). *For $H \notin \Sigma_\lambda$,*

$$f_\lambda(H) = \min_{\substack{P \in \mathbb{R}^b[x], \deg P \leq t-1 \\ \|P(0)\|_2=1}} \left(\sum_{i=1}^m [h_i^T P(\lambda_i)]^2 \right)^{1/2}. \quad (5.2)$$

The minimum is attained. If H has the μ, R bounds of Lemma 4.1, there is a matrix $B \in \mathbb{R}^{m \times b}$ with $\|B\|_F = 1$ such that

$$\limsup_{s \downarrow 0} \frac{f_\lambda(H + sB) - f_\lambda(H)}{s} \leq -\alpha, \quad \alpha = \frac{\mu\gamma\rho}{4\sqrt{m}R}. \quad (5.3)$$

Proof. The invertible linear map $K_\lambda(H)$ identifies coefficient vectors with data $y_i = h_i^T P(\lambda_i)$; the constant coefficient is $E_{\lambda,H}(0)y$. Thus the minimum in (5.2) is $\min\{\|y\|_2 : \|E_{\lambda,H}(0)y\|_2 = 1\} = 1/\|E_{\lambda,H}(0)\|_2$. A right singular vector for the largest singular value attains this minimum. In particular, there is no compactness assumption on an unbounded polynomial coefficient set.

Let P attain the minimum and set

$$y_i = h_i^T P(\lambda_i), \quad f = f_\lambda(H), \quad u = y/f.$$

Varying the positive-degree coefficient vectors of P freely leaves $P(0)$ fixed. First-order optimality of the squared residual gives $H^T \Lambda^j u = 0$ for $1 \leq j \leq t-1$. Put $a = P(0)$ and $\phi(x) = a^T P(x)$. Then $\|a\|_2 = 1$, $\phi(0) = 1$, and $|\phi(\lambda_i)| \leq \|P(\lambda_i)\|_2$. Lemma 4.1 therefore implies

$$D := \left(\sum_i u_i^2 \|P(\lambda_i)\|_2^2 \right)^{1/2} \geq \alpha. \quad (5.4)$$

Let row i of B be $-u_i P(\lambda_i)^T / D$. Then $\|B\|_F = 1$. Keep this particular P fixed when replacing H by $H + sB$. It remains feasible for (5.2); its residual norm is differentiable at $s = 0$ because $f > 0$. Its derivative is

$$\sum_i u_i B_i^T P(\lambda_i) = -D \leq -\alpha.$$

Since $f_\lambda(H + sB)$ is no greater than this feasible residual norm and equality holds at $s = 0$, (5.3) follows. This argument does not require a unique minimizing polynomial or a simple largest singular value of $E(0)$. $\square$

Lemma 5.2 (A local error bound). *Fix $H_0 \notin \Sigma_\lambda$ and $r_0 > 0$. Suppose every matrix X in the closed Frobenius ball $\bar{B}_F(H_0, r_0)$ has spectral norm at most R and contiguous b -row singular values at least μ . With α as in (5.3),*

$$f_\lambda(H_0) \geq \frac{\alpha}{2} \min\{r_0, \text{dist}_F(H_0, \Sigma_\lambda)\}. \quad (5.5)$$

Proof. Suppose the strict reverse inequality holds, write $f_0 = f_\lambda(H_0)$, and put $r = 2f_0/\alpha$. Then $r < r_0$ and $r < \text{dist}_F(H_0, \Sigma_\lambda)$. The function

$$g(X) = f_\lambda(X) + \frac{\alpha}{2} \|X - H_0\|_F$$

is continuous on the compact closed radius- r ball, which is disjoint from Σ_λ . On its boundary, $g(X) = f_\lambda(X) + f_0 > f_0 = g(H_0)$. Its minimum is therefore attained in the interior, at some X_* . Lemma 5.1 gives a unit direction with upper directional derivative of f_λ at most $-\alpha$. The distance penalty has directional increment at most $\alpha/2$ times the step, by the triangle inequality. Thus g has a strictly decreasing direction at X_* , a contradiction. $\square$

Lemma 5.2 is the bridge between an extrapolation norm and algebraic geometry. It does not assert that a raw Krylov singular value equals distance to the starting-matrix singular locus. Such an identification would generally be false.

6 A probability bound uniform in the head spectrum

We use the following consequence of a published real-algebraic tube theorem. The ambient space is real, as is the Gaussian starting block.

Lemma 6.1 (Imported conic tube bound). *Let $\Sigma \neq \{0\}$ be the zero set of a nonzero homogeneous real polynomial of degree $d_0 \geq 1$ on $\mathbb{R}^{p+1}$, $p \geq 1$. For a standard Gaussian $g \in \mathbb{R}^{p+1}$, define $\mathcal{C}_\Sigma(g) = \|g\|_2 / \text{dist}(g, \Sigma)$. Then, for $v \geq 3d_0p$,*

$$\mathbb{P}\{\mathcal{C}_\Sigma(g) \geq v\} \leq \frac{40d_0p}{v}. \quad (6.1)$$

Derivation from the imported theorem. Theorem 1.1 of Bürgisser, Cucker, and Lotz [4], in its average-case specialization, bounds the probability by

$$4 \sum_{j=1}^{p-1} \binom{p}{j} (2d_0)^j (1 + v^{-1})^{p-j} v^{-j} + \frac{2pO_p}{O_{p-1}} (2d_0/v)^p, \quad (6.2)$$

where O_p is the surface area of S^p . Their statement permits singular real algebraic sets. The Gaussian direction is uniform on S^p , and the condition number is homogeneous of degree zero, so the spherical estimate applies unchanged.

The sum in (6.2) is at most $4[\exp(3d_0p/v) - 1] \leq 24d_0p/v$, since $v \geq 3d_0p$ and $e^z - 1 \leq 2z$ for $0 \leq z \leq 1$. Also $O_p/O_{p-1} = \int_0^\pi \sin^{p-1} \theta d\theta < 4$ and $2d_0/v \leq 1$, so the last term is at most $16d_0p/v$. Adding proves (6.1). Only (6.2) is imported; this simplification is included to make the constants explicit. $\square$

For completeness, the elementary Gaussian estimates needed with it are proved here.

Lemma 6.2 (Gaussian row-block estimates). *If $g \sim N(0, 1)$, $H \in \mathbb{R}^{m \times b}$ is standard Gaussian, and $S \in \mathbb{R}^{b \times b}$ is standard Gaussian, then for $a > 0$,*

$$\mathbb{P}\{|g| < a\} \leq a, \quad \mathbb{E}\|H\|_F^2 = mb, \quad \mathbb{P}\{\sigma_{\min}(S) < a\} \leq b^{3/2}a. \quad (6.3)$$

Proof. The scalar bound follows by integrating the normal density, which is at most $1/\sqrt{2\pi}$, over an interval of length $2a$. The expectation is the sum of the entry variances. For the last bound, let d_i be the distance from row i of S to the span of its other rows. Almost surely S is invertible and the norm of column i of S^{-1} is d_i^{-1} . Hence

$$\|S^{-1}\|_2 \leq \|S^{-1}\|_F \leq \sqrt{b} \max_i d_i^{-1}.$$

Conditioned on the other rows, d_i has distribution $|g|$. If $\sigma_{\min}(S) < a$, at least one $d_i < \sqrt{b}a$. The union bound proves the claim. No independence among the row-distance events is required. $\square$

Theorem 6.3 (Anchored extrapolation). *For every admissible positive head spectrum, standard Gaussian $H \in \mathbb{R}^{m \times b}$, and $0 < \eta < 1/2$,*

$$\mathbb{P} \left\{ \|E_{\lambda, H}(0)\|_2 \leq 2^{24} m^8 \eta^{-4} \left(\frac{128e}{\Delta} \right)^t \right\} \geq 1 - \eta. \quad (6.4)$$

For $t \geq 2$, the bound holds on an explicitly specified event depending on the spectrum only through Σ_λ and the stated lower gap bound Δ .

Proof. First take $t \geq 2$, so $m \geq 2$ and $N = mb \geq 2$. Put $p = N - 1$ and

$$R_0 = 2\sqrt{N/\eta}, \quad \tau = \frac{\eta}{4mb^{3/2}}.$$

Consider the simultaneous event

$$\|H\|_F \leq R_0, \quad (6.5)$$

$$\sigma_{\min}(H_{\{r, \dots, r+b-1\}}) \geq \tau \quad (1 \leq r \leq m - b + 1), \quad (6.6)$$

$$\|H\|_F \geq \eta/4, \quad (6.7)$$

$$\frac{\|H\|_F}{\text{dist}_F(H, \Sigma_\lambda)} \leq \frac{160mp}{\eta}. \quad (6.8)$$

Each failure probability is at most $\eta/4$. For (6.5), use Markov's inequality and $\mathbb{E}\|H\|_F^2 = N$. For (6.6), apply (6.3) to at most m contiguous b -row blocks. For (6.7), the event of a smaller Frobenius norm implies $|H_{11}| < \eta/4$. For (6.8), use Lemma 6.1 with $d_0 = m$ and Lemma 2.1. The cone contains nonzero matrices with a zero row, and it is not the whole ambient space. Thus every hypothesis of the imported theorem holds. A union bound gives probability at least $1 - \eta$.

On this event,

$$\text{dist}_F(H, \Sigma_\lambda) \geq \frac{\eta^2}{640mp} \geq \frac{\eta^2}{640m^3} =: d_*. \quad (6.9)$$

Here $p \leq mb \leq m^2$. In the closed ball of radius $r_0 = \tau/2$ around H , perturbation of a smallest singular value by at most the perturbation norm gives row-block lower bound

$$\mu = \tau/2 \geq \frac{\eta}{8m^{5/2}}.$$

Every matrix in the ball has spectral norm at most $R = 2R_0 \leq 4m/\sqrt{\eta}$; indeed $r_0 < R_0$. Therefore Lemmas 4.1 and 5.2 apply throughout this ball, with

$$\alpha = \frac{\mu\gamma\rho}{4\sqrt{m}R} \geq \frac{\eta^{3/2}\Delta\rho}{8192m^5}, \quad \rho = \left(\frac{\Delta}{128e} \right)^{t-1}. \quad (6.10)$$

Also $r_0 \geq \eta/(8m^{5/2}) \geq d_*$. Consequently

$$\begin{aligned} f_\lambda(H) &\geq \frac{\alpha}{2} d_* \geq \frac{\eta^{7/2} \Delta \rho}{10485760 m^8} \\ &\geq \frac{\eta^4}{2^{24} m^8} \left(\frac{\Delta}{128e} \right)^t. \end{aligned} \quad (6.11)$$

The last step only weakens the bound: $10485760 < 2^{24}$, $\eta^{7/2} \geq \eta^4$, and $\Delta \rho \geq (\Delta/(128e))^t$. Inverting gives (6.4).

When $t = 1$, $m = b$ and $E_{\lambda,H}(0) = H^{-1}$, independently of the nodes. Lemma 6.2 gives $\|H^{-1}\|_2 \leq m^{3/2}/\eta$ with probability at least $1 - \eta$. This is smaller than the right-hand side of (6.4), with $\Delta = 1$. It also handles the one-dimensional case $m = b = 1$, for which the conic tube theorem was not invoked. $\square$

Theorem 6.4 (Uniform tail interpolation). *Under the same assumptions,*

$$\mathbb{P} \left\{ \sup_{0 \leq x \leq \lambda_m} \|E_{\lambda,H}(x)\|_2 \leq 2^{24} m^8 \eta^{-4} \left(\frac{128e}{\Delta} \right)^t \right\} \geq 1 - \eta. \quad (6.12)$$

Proof. For $t \geq 2$, use the single event (6.5)–(6.8) for the original spectrum. For every $0 \leq x < \lambda_m$, all shifted nodes $\lambda_i - x$ are positive and

$$\frac{(\lambda_i - x) - (\lambda_{i+b} - x)}{\lambda_i - x} \geq \frac{\lambda_i - \lambda_{i+b}}{\lambda_i} \geq \Delta.$$

By Lemma 2.2, their singular locus is *exactly* Σ_λ . Thus the same four events, local ball, and estimates (6.9)–(6.11) apply to every shifted spectrum with the same constants. This is a deterministic simultaneous implication on that event, not an uncountable union of probability claims.

The evaluation identity in (2.4) proves the bound for every $x < \lambda_m$. Since $E_{\lambda,H}(x)$ is a matrix polynomial, continuity gives it also at $x = \lambda_m$. For $t = 1$, $E(x) = H^{-1}$ is constant and the assertion follows from Theorem 6.3. $\square$

This establishes the sufficient interpolation estimate left open in the earlier reports. The dimension and failure probability appear only as fixed polynomial factors; their logarithms are not multiplied by t .

7 Transfer to the original randomized algorithm

We now return to $M = AA^T$ and the original G . Take a fixed orthogonal eigenbasis $U = [U_m \ U_\perp]$ and write

$$U^T M U = \text{diag}(\Lambda, \Lambda_\perp), \quad U^T G = \begin{bmatrix} H \\ T \end{bmatrix}.$$

Orthogonal invariance gives standard Gaussian entries in both blocks. No eigenvectors are selected using G .

Let

$$B = [G, MG, \dots, M^{t-1}G], \quad K_T = [T, \Lambda_\perp T, \dots, \Lambda_\perp^{t-1}T].$$

Almost surely,

$$BK_\lambda(H)^{-1} = U \begin{bmatrix} I_m \\ F \end{bmatrix}, \quad F = K_T K_\lambda(H)^{-1}. \quad (7.1)$$

At a tail node $x_j \in [0, \lambda_m]$, row j of F is $T_j^T E_{\lambda, H}(x_j)$. Hence

$$\|F\|_2^2 \leq \|F\|_F^2 \leq \|T\|_F^2 \sup_{0 \leq x \leq \lambda_m} \|E_{\lambda, H}(x)\|_2^2. \quad (7.2)$$

Apply Theorem 6.4 with $\eta = \delta/2$, and intersect its event with $\|T\|_F^2 \leq 2nb/\delta$. The latter fails with probability at most $\delta/2$ by Markov's inequality, since $\mathbb{E}\|T\|_F^2 = (n-m)b \leq nb$. No independence is needed for this intersection. We obtain, with probability at least $1 - \delta$,

$$L := 1 + \|F\|_2^2 \leq 1 + 2^{57} n b m^{16} \delta^{-9} \left(\frac{128e}{\Delta} \right)^{2t} \leq 1 + 2^{57} n^{18} \delta^{-9} \left(\frac{128e}{\Delta} \right)^{2t}. \quad (7.3)$$

For clarity, the graph-to-overlap reduction is proved rather than assumed. Let $J_k \in \mathbb{R}^{m \times k}$ contain the first k coordinate vectors and let $F_k = F J_k$. Then

$$Q = U \begin{bmatrix} J_k \\ F_k \end{bmatrix} (I_k + F_k^T F_k)^{-1/2}$$

has orthonormal columns in $\text{range}(B)$, and

$$\|(U_k^T Q)^{-1}\|_2^2 = \|I_k + F_k^T F_k\|_2 \leq L. \quad (7.4)$$

This handles k not divisible by b directly.

Theorem 7.1 (Imported deterministic good-start convergence). *There is an absolute $C_0 \geq 1$ with the following property. If the column space of a starting matrix B contains orthonormal $Q \in \mathbb{R}^{n \times k}$ satisfying (7.4), then projection onto $\mathcal{K}_s(M, B)$ followed by a rank- k SVD truncation satisfies (1.4)–(1.5) for sufficiently many steps,*

$$s \geq \max \left\{ 2, \left\lceil \frac{C_0}{\sqrt{\varepsilon}} \log \frac{nL}{\varepsilon} \right\rceil \right\}.$$

This is Imported Theorem 3.2 in Chen, Epperly, Meyer, Musco, and Rao [2], with its sufficient-iteration $O(\cdot)$ notation expressed using an absolute constant. Their Problem 1.1 explicitly specifies *right* singular vectors. The theorem invokes the good-start analysis in Appendix F of Meyer, Musco, and Musco [3]. It is the only convergence theorem imported here. The case $\lambda_{k+1} = 0$ is also proved directly below rather than relying on a zero-denominator limit in that analysis.

Proof of Theorem 1.1. Suppose first $\lambda_{k+1} > 0$. On the probability- $1 - \delta$ event above, (7.3) and (7.4) hold. Direct multiplication of the block powers gives the exact identity

$$\mathcal{K}_s(M, B) = \mathcal{K}_{s+t-1}(M, G). \quad (7.5)$$

Put $X = t \log(2/\Delta)$ and $Y = \log(n/(\delta\varepsilon))$. The bound (7.3) implies

$$\begin{aligned} \log \frac{nL}{\varepsilon} &\leq 58 \log 2 + 19 \log n + 9 \log(1/\delta) + \log(1/\varepsilon) + 2t \log(128e/\Delta) \\ &\leq 80(X + Y). \end{aligned}$$

For the last inequality, use $58 \log 2 \leq 58X$, $\log(128e/\Delta) \leq 9 \log(2/\Delta)$, and $19 \log n + 9 \log(1/\delta) + \log(1/\varepsilon) \leq 19Y$. Consequently a choice such as $C = 80C_0 + 4$ in (1.3) ensures $q - t + 1 \geq \max\{2, \lceil C_0 \log(nL/\varepsilon)/\sqrt{\varepsilon} \rceil\}$. Theorem 7.1 and (7.5) prove all three guarantees for the actual output of (1.2).

If $\lambda_{k+1} = 0$, then $\text{rank}(A) = k$ and the hypothesis $m \leq \text{rank}(A)$ forces $m = k$, so b divides k . In the chosen eigenbasis,

$$U^T[MG, M^2G, \dots, M^tG] = \begin{bmatrix} \Lambda K_\lambda(H) \\ 0 \end{bmatrix}.$$

Both head factors are invertible almost surely. Thus $\mathcal{K}_{t+1}(M, G)$ contains $\text{range}(A)$, giving $ZZ^T A = A$ and an exact best rank- k approximation. The right singular-vector energies are exactly λ_i . The above choice of C ensures $q \geq t + 1$, so this case is included as well. $\square$

8 Scope, failure modes avoided, and review obligations

The proof is uniform in all of $n, d, k, b, t, \varepsilon, \delta$ and in the positive head spectrum. It places no bound on λ_1/λ_m and requires no positive consecutive gap. It leaves the matrix A and Gaussian distribution in RA-04 unchanged. The word “perturbation” in the cited tube theorem does not denote an extra algorithmic perturbation here: only its average-case spherical specialization is used.

Several potential shortcuts are deliberately not used. The determinant being nonzero does not alone give a quantitative bound; the descent lemma and the tube theorem supply that missing step. Smallest singular values of raw monomial Krylov matrices are not asserted to be gap-only. Noncommuting matrix-valued Lagrange products are not introduced. Finally, the uniform evaluation bound follows from a common event and exact translation invariance, not from an uncountable union bound.

Proof obligation	Disposition
Repeated head values	They occur at most b times under the gap assumption; the residue-class witness and annular row groups allow repetitions.
Unaligned or wide clusters	Only the b -step relative gap and contiguous row blocks are used. No fixed cluster centers or width condition occur.
Roots at head nodes	Contour boundaries avoid all root moduli in the covering. Roots inside a contour do not invalidate the boundary identity.
Complex roots	The reverse triangle inequality gives a bound on an entire circle from the root modulus.
Multiple extremal singular values	The descent proof freezes one minimizer and uses an upper directional derivative, not differentiability of the minimum.
Singular algebraic locus	The imported tube theorem covers singular real algebraic sets. No smoothness of the determinant-zero set is imposed.
Probability dependence	Four elementary/conic events are combined by a union bound. Their independence is not asserted.
All tail evaluations	The determinant-zero locus is invariant under common node translations. One event controls all $x < \lambda_m$; continuity gives the endpoint.
Nondivisible target rank	The first k graph columns give the overlap certificate directly.
Zero tail error	Exact range recovery at depth $t + 1$ is proved separately.

Proof obligation	Disposition
Right-vector energy guarantee	Imported Theorem 3.2 is used with the right-vector convention of its Problem 1.1, not a substituted left-vector estimate.
Universal constants	A fixed polynomial prefactor and exponent- t gap factor give (7.3); $C = 80C_0 + 4$ is one sufficient choice relative to the imported constant.

Evidence level. The complete argument passed a separate Codex AI-agent informal mathematical audit. Under the repository’s policy this supports *Solved* [5]. The independent report accompanies this submission. No external human peer review, formal proof certificate, Lean verification or novelty-priority claim is asserted.

The independent audit examined the annular identity (4.6), the passage from a frozen minimizing polynomial to the descent inequality (5.3), and the compact-ball argument in Lemma 5.2. These are the new steps connecting spectral clustering to the published algebraic probability theorem. Their assumptions and proofs are stated in full above; they are not outstanding conjectural premises.

9 Reproducible checks

The companion scripts are tests of formulas and implementations, not substitutes for the universal arguments. Rational fixtures are expressly not presented as Gaussian samples. Floating-point observations are not probability certificates.

Exact checks. The rational-arithmetic suite passes five residue-class rank witnesses, including two $t = 1$ endpoints, and three interpolation fixtures. It verifies six vector resolvent identities, 105 positive-pole coordinate residues, translation and determinant invariance, Schur-complement formulas for the anchored minimum, and direct graph identities. Four exact interval-cover fixtures include repeated logarithmic root locations and verify the total-length bound. Additional checks verify 13 symbolic Pascal translation identities and a tied-singular-value fixture in which the frozen-minimizer descent remains valid without uniqueness.

Head-matrix diagnostics. Across 96 fixtures with $b, t \in \{1, 2, 3, 4\}$, geometric, repeated, and nonzero-width head spectra were tested. The largest relative discrepancy between the analytic singular-value derivative and the frozen-minimizer derivative was $6.05 \cdot 10^{-14}$; the corresponding finite-difference discrepancy was $6.75 \cdot 10^{-5}$. The maximum normalized moment residual was $4.03 \cdot 10^{-16}$, and the maximum relative translation residual was $8.10 \cdot 10^{-13}$. The stated interpolation bound exceeded every sampled norm, but its conservatism and these finite observations do not establish a probability bound.

Algorithmic diagnostics. Two-pass block Arnoldi with numerical-rank detection in binary64 produced 81 records across seven spectra and three seeds per spectrum. Every record directly measures both norm errors and the ordered right-singular-vector energy defect. At $\varepsilon = 0.1$, 43 records satisfied all tested criteria. The table gives the smallest *tested* depth at which all three seeds passed. Failed smaller-depth records are retained. Neither the table nor its success counts estimate or certify the universal constant in Theorem 1.1.

Fixture	n	b	k	Tested depth
Exactly repeated clusters	72	4	24	18
Wide, unaligned spectrum	64	4	19	10
Mixed multiplicities at the boundary	42	3	10	9
Nonzero-width clusters	45	3	14	10
Scalar starting block	20	1	5	12
Full-size starting block	32	6	6	4
Zero optimal error	40	3	12	5

For the zero-error fixture, “passed” means that the spectral residual, Frobenius residual, and energy defect are each at most 10^{-10} ; the theorem is an exact-arithmetic assertion, not this tolerance test. Its exact recovery depth is $t + 1 = 5$.

Spectral-scale stress tests. Six further interpolation solves used 180-decimal-digit arithmetic. For the scalar fixture with head spectrum $(1, \zeta, \zeta/2)$ and $\zeta = 10^{-4}, 10^{-16}, 10^{-50}$, the clustered gap stayed at $1/2$. The raw smallest singular values were approximately $1.65 \cdot 10^{-5}$, $1.65 \cdot 10^{-17}$, and $1.65 \cdot 10^{-51}$, while the anchored evaluation norms were approximately 4.07350, 4.07329, and 4.07329. This finite diagnostic illustrates the difference between raw conditioning and extrapolation; it does not prove a limiting statement. Normalized inverse residuals are recorded as backward residual diagnostics, not forward-error certificates.

The recorded environment is Python 3.13.5, NumPy 2.3.5, SymPy 1.14.0, and mpmath 1.3.0. Platform-dependent floating-point variation is expected.

From the package root, run:

```
python -m pip install -r requirements.txt
python tests/exact_checks.py --output results/exact_checks.json
python tests/numerical_checks.py --output-dir results
./build.sh
```

The tests retain seeds, dimensions, numerical residuals, and failed smaller-depth approximation criteria. Identity-check failures stop execution. The source and a machine-readable source inventory are included. The prior packages are identified by hashes in the provenance file; no earlier unproved interpolation estimate is assumed by this proof.

References

- [1] A. Townsend, *Open Problems in Numerical Linear Algebra*, entry RA-04, “Clustered singular-value gaps in randomized block Krylov approximation.” Repository statement consulted September 12, 2026 (US Eastern time). <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/RA-04>.
- [2] T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao, *Does block size matter in randomized block Krylov low-rank approximation?* SODA 2026, pp. 1026–1046. doi:10.1137/1.9781611978971.42. Version used: arXiv:2508.06486v2, Problem 1.1 and Imported Theorem 3.2, PDF pp. 1 and 6. The PDF title page identifies Christopher Musco and Akash Rao.
- [3] R. A. Meyer, C. Musco, and C. Musco, *On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation*, SODA 2024, pp. 811–845. arXiv:2305.02535. Appendix F is the convergence-transfer provenance cited by [2].
- [4] P. Bürgisser, F. Cucker, and M. Lotz, *The probability that a slightly perturbed numerical analysis problem is difficult*, Mathematics of Computation 77 (2008), 1559–1583. doi:10.1090/S0025-5718-08-02060-7. Theorem 1.1 in the inspected preprint arXiv:math/0610270v1, titled *The probability that a small perturbation of a numerical analysis problem is difficult*, PDF p. 4. The real theorem, not the separate complex conic theorem, is used.
- [5] *Open Problems in Numerical Linear Algebra*, README, “Problem status,” and CONTRIBUTING.md. Consulted September 12, 2026 (US Eastern time). <https://github.com/ajt60gaibb/OpenProblemsInNLA>.